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INDUSTRY ELECTIVE - POWER BI

Sales Performance Analysis Dashboard using Power BI

Problem Statement

Organizations generate large volumes of sales data across different regions, customer segments, and product categories. However, analyzing this data manually makes it difficult to identify trends, top-performing products, and customer behavior patterns.
This project aims to develop an interactive Sales Performance Dashboard using Power BI to transform raw sales data into meaningful insights that support better business decision-making.

Objective of the Project

The main objective of this project is to design and develop an interactive dashboard that:

- Analyzes overall sales performance
- Identifies region-wise sales distribution
- Evaluates category and sub-category performance
- Detects top-performing products
- Examines customer segment contributions
- Tracks sales trends over time
- Enables interactive filtering using slicers

This helps stakeholders understand business performance efficiently.

Dataset Description

The dataset used in this project contains retail sales transaction details including:

- Order ID
- Order Date
- Customer ID
- Customer Name

- Segment
- Region
- State
- Category
- Sub-Category
- Product Name
- Ship Mode
- Sales

The dataset represents sales activity across multiple regions of the United States and supports detailed performance analysis.

Tools and Technologies Used

The following tools were used to develop this project:

- Microsoft Power BI Desktop
- CSV Dataset (Superstore Sales Data)
- Data Cleaning using Power Query
- DAX Measures for calculations
- Interactive Visualizations

Dashboard Features

The dashboard consists of four main analytical pages:

1. Overview Dashboard

This page provides a summary of overall sales performance using KPI indicators and category-level insights.

Features include:

- Total Sales KPI
- Total Orders KPI
- Total Customers KPI
- Average Sales KPI
- Sales by Region
- Sales by Customer Segment
- Sales by Category

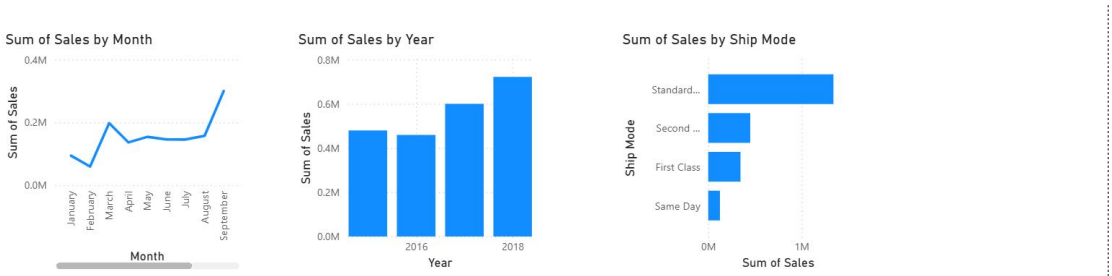
2. Sales Trend Analysis

This page analyzes how sales performance changes over time.

Features include:

- Monthly Sales Trend
- Year-wise Sales Performance
- Sales by Shipping Mode

These visuals help identify seasonal trends and yearly growth patterns.



3. Product Insights Dashboard

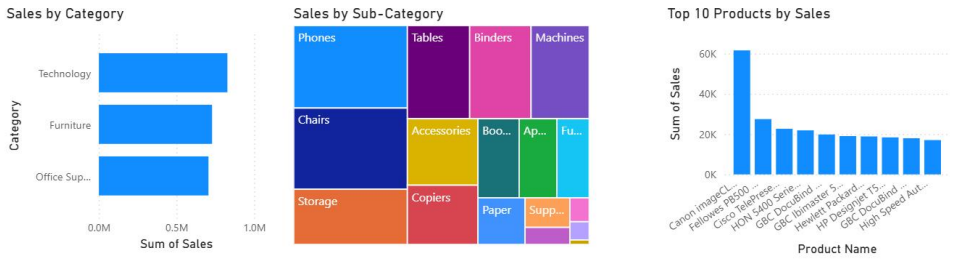
This page highlights product-level performance.

Features include:

- Sales by Category
- Sales by Sub-Category

- Top 10 Products by Sales

This helps identify high-performing products and product groups.



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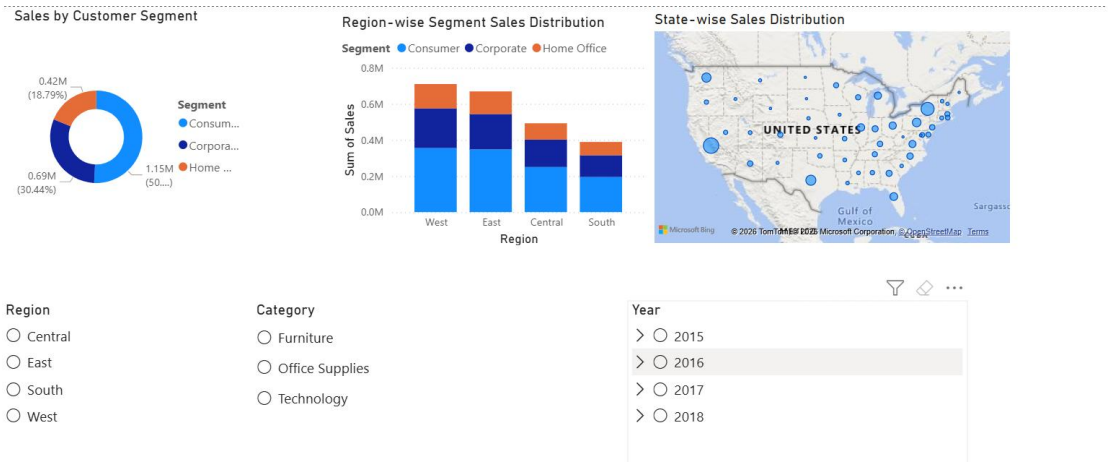
4. Customer Analysis Dashboard

This page analyzes customer contribution and geographical distribution.

Features include:

- Sales by Customer Segment
- Region-wise Segment Sales Distribution
- State-wise Sales Distribution Map
- Interactive Slicers (Region, Category, Year)

This allows dynamic filtering and deeper business insights.



Key Insights from the Dashboard

Several important business insights were observed:

- The West region generated the highest overall sales.
- Consumer segment contributed the largest share of revenue.
- Technology category recorded the highest sales among product categories.
- Standard shipping mode was used most frequently.
- Certain products contributed significantly higher revenue compared to others.
- Sales showed noticeable variation across different years and months.

These insights help identify business strengths and improvement areas.

Unique Features of the Project

This dashboard includes several advanced Power BI features such as:

- Interactive KPI Cards
- Hierarchical Date Analysis
- Top N Product Filtering
- Treemap Visualization
- Geographic Map Visualization
- Stacked Comparison Charts
- Dynamic Slicers for Filtering
- Multi-page Analytical Dashboard Design

These features improve usability and decision-making efficiency.

Future Improvements

This project can be further enhanced by:

- Adding profit analysis for better financial insights
- Integrating real-time sales data sources
- Applying predictive analytics for future sales forecasting
- Creating role-based dashboards for different departments
- Publishing dashboard to Power BI Service for cloud access

These improvements would make the dashboard more scalable and business-ready.